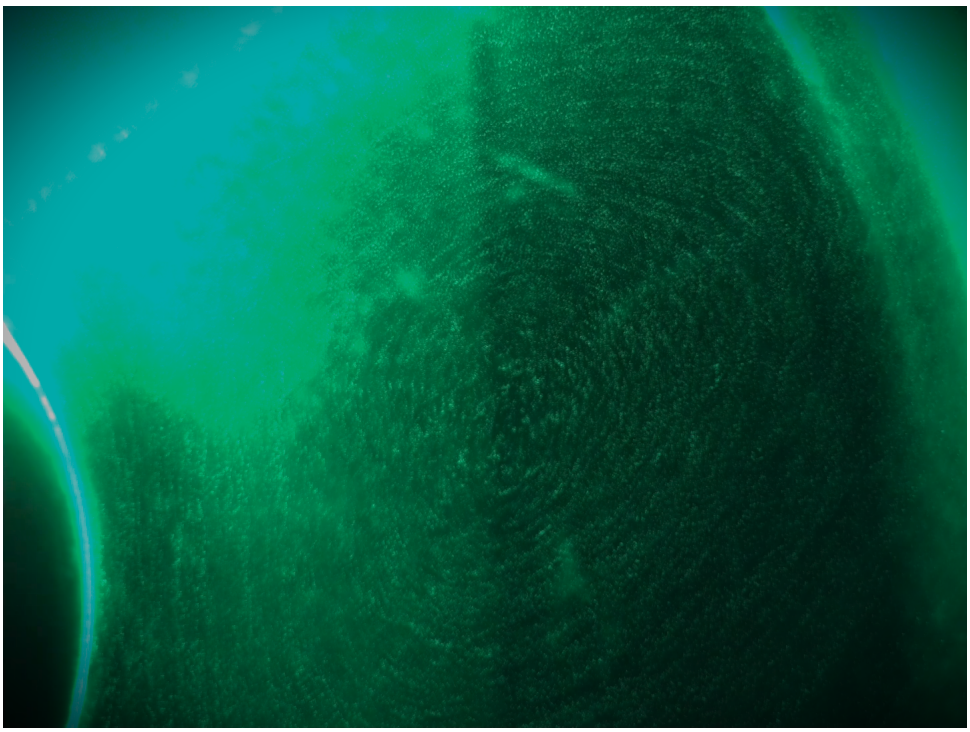
University of Toronto

ADVANCED PHYSICS LABORATORY

ATM

Weather in a Tank



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1 Introduction

Large-scale atmospheric circulation plays a vital role in transporting heat and moisture, shaping weather patterns essential for life on Earth. Fluid dynamics provide a powerful tool for understanding the phenomena governing these flows. However, the fluid systems described by these dynamical equations are complex, covering a wide range of qualitatively different behaviours. At the planetary scale, many common weather patterns (for example, the mid-latitude westerlies, subtropical ocean gyres, and near-surface rainfalls) belong to a subset of flows which can be well approximated with the balance of two or more dominant forces. These are commonly referred to as *balanced flows*. By restricting our attention to these simpler cases, we are able to study their analogues in scaled-down laboratory setting.

Many of Earth's atmospheric signatures are driven by its rate of rotation and received solar heating. By investigating the interplay between these two sources and how they affect the fluid's velocity field through a series of experiments, you will be able to piece together how differential heating and rotation generate the distinct weather patterns we see on Earth. Additionally, by testing observations against their simplified mathematical models you will test the validity of the *balanced flow* assumption and gain intuition behind distinct regimes of flow.

This experiment will use a spinning tank filled with a column of water to simulate the Earth's atmosphere. You will be able to vary the temperature gradient across the column by placing an ice bath at the centre of the tank. The effect of rotation can be adjusted with a fine control turn table. Data on the winds generated (flow field) will be collected through a technique called particle image velocimetry (PIV). A horizontal laser sheet will be used to illumiate microparticles in the water and imaged by an overhead camera.

2 Theory

The theoretical framework used to model atmospheric flows are generally set by the equations of motion known as the *primitive equations*. The equations come from 1 conservation of momentum, 2 conservation of mass, and 3 conservation of energy. For a full derivation, see Reference [7].

$$\frac{\mathrm{D}}{\mathrm{D}t}\mathbf{u} = -\frac{1}{\rho}\nabla p + \mathbf{g_{tot}} - f\hat{\mathbf{z}} \times \mathbf{u} + \frac{1}{\rho}\frac{\partial \tau}{\partial z} \qquad \text{3D Inviscid Momentum Balance} \tag{1}$$

$$\frac{\mathrm{D}}{\mathrm{D}t}\rho + \rho\vec{\nabla} \cdot \mathbf{u} = 0 \qquad \text{Continuity Equation} \tag{2}$$

$$\frac{\mathrm{D}}{\mathrm{D}t}T = -\frac{p}{\rho c_v}\nabla \cdot \mathbf{u}_h + \frac{J}{\rho c_v} \qquad \text{Conservation of Energy} \tag{3}$$

Where:

- $\mathbf{u}(\mathbf{x}, t) = u\hat{\mathbf{x}} + v\hat{\mathbf{y}} + w\hat{\mathbf{z}}$ is a 3-dimensional vector describing the velocity of the fluid at a certain point and at a certain time
- $p(\mathbf{x}, t)$ is the pressure
- $\rho(\mathbf{x}, t)$ is the density of the fluid
- $\mathbf{g_{tot}}$ is a force acting along the surface normal $\hat{\mathbf{z}}$ accounting for gravity and centripetal acceleration in the rotating frame
- $f = 2\Omega \sin\phi$ is the Coriolis parameter under the f-plane approximation (Reference [7]). $\Omega \sin\phi$ denotes the vertical component of Earth's angular velocity
- τ is the prescribed frictional stress
- $T(\mathbf{x}, t)$ is the temperature of the fluid
- and J is the prescribed heating

Typically, this set of equations is also paired with an equation of state to close the system:

$$f(p, \rho, T) = 0.$$

Such as in the case of an ideal gas, $p = \rho RT$. Since the set of equations are nonlinear and dissipative, qualitative behaviour can vary drastically depending on the flow regime and in general, closed-form solutions are not always possible.

A *balanced flow* describes a case where two or more forces are dominant in the momentum equation. In these limiting cases, we can write simpler approximations of Equation 1 for analytical insight. In the following section, we use scaling to argue the theoretical validity of various balanced flows.

2.1 Scale Analysis

Since we are mainly interested in large-scale motions in the atmosphere, we will investigate the relative sizes of the terms in the primitive equations. We scale Equation 1 by setting a characteristic horizontal length scale L, a characteristic vertical length scale H, as well as characteristic velocities for horizontal U and vertical flow W. We will also introduce f_o which is a typical mid latitute frequency for the Coriolis parameter. The nondimensional horizontal wind is written $\tilde{\mathbf{u}}_h = \frac{1}{U}\mathbf{u}_h$, nondimensional time $\tilde{t} = \frac{U}{L}t$, and nondimensional horizontal space as $(\tilde{x}, \tilde{y}) = \frac{1}{L}(x, y)$, nondimensional height $\tilde{z} = \frac{1}{H}z$. Also, $\tilde{f} = \frac{1}{f_o}f$ for an average f_o Coriolis frequency in the mid-latitudes, making nondimensional pressure $\tilde{p} = \frac{1}{f_o UL}p$. Since the coriolis force only acts on horizontal flows in our f-plane approximation, the horizontal components reduce to:

$$Ro\left(\frac{\partial}{\partial\tilde{t}} + \tilde{\mathbf{u}}_h \cdot \tilde{\nabla}_h + \frac{WL}{UH}\tilde{u_z}\frac{\partial}{\partial\tilde{z}}\right)\tilde{\mathbf{u}}_h = -\frac{1}{\rho}\tilde{\nabla}_{\tilde{\mathbf{z}}}p - \tilde{f}\hat{\tilde{\mathbf{z}}} \times \tilde{\mathbf{u}}_h + \frac{1}{\rho}\frac{\partial\tilde{\tau}}{\partial\tilde{\mathbf{z}}} \tag{4}$$

Where $Ro = \frac{U}{f_o L}$ is the nondimensional Rossby number. In the atmosphere, typical values for these motions are $U \approx 10m/s$ and $L \approx 10^6 m$, $f_o \approx 10^{-4} Hz$ for mid-latitudes; the vertical velocity coming from thermal forcing is around $W \approx 10^{-3} m/s$ and the length scale is given by the hydrostatic and ideal gas approximation to be $10^3 m$ [5].

In this range, $Ro \approx 10^{-1}$, meaning the horizontal acceleration scales by 10^{-1} and the vertical variation in velocilty scales like $Ro\frac{WL}{UH} \approx 10^{-3}$ while the coefficients for the forces on the right are of order unity. For balanced flows, Equation 4 reduces to Equation 5.

$$0 \approx -\frac{1}{\rho}\tilde{\nabla}_{\tilde{\mathbf{z}}}p - \tilde{f}\hat{\tilde{\mathbf{z}}} \times \tilde{\mathbf{u}}_h + \frac{1}{\rho}\frac{\partial\tilde{\tau}}{\partial\tilde{\mathbf{z}}} \tag{5}$$

2.2 Geostrophic Wind

For flows far away from frictional surfaces such as at mid to high altitudes, the frictional term can be ignored in Equation 5 and the geostrophic force will balance the pressure gradient and produce a fixed, horizontal flow along isobars (lines of constant pressure) 6.

$$\tilde{f}\hat{\tilde{\mathbf{z}}} \times \tilde{\mathbf{u}}_h \approx -\frac{1}{\rho}\tilde{\nabla}_{\tilde{\mathbf{z}}}p \tag{6}$$

$$\tag{7}$$

2.2.1 Inertial Flows

If we write the horizontal momentum equations in polar coordinates, it becomes evident that a balance of forces perpendicular to the flow also includes a radial force term from centripetal acceleration. If the pressure is uniform along a reference surface, and we neglect friction, then the Coriolis force can balance the centripetal force in Equation 8.

$$\frac{\tilde{u}_\phi^2}{\tilde{R}} + \tilde{f}\tilde{u}_\phi \approx 0 \tag{8}$$

Where $\tilde{u}_\phi$ is the velocity in the azimuthal direction and $\tilde{R}$ is the radius of the flow. These fixed circular flows are often found above oceans where the currents are set by transient winds and not internal pressure gradients. Equation 8 gives us a relation between the size of the concentric flow and the velocity. It is then possible to image the velocity and radius of surface tracers to verify the validity of this relation and compute the period of the motion, relating it to a latitudinal gradient in intertial winds by varying f.

2.3 Taylor Columns

To see how the geostrophic wind varies in the vertical direction, we take the curl of Equation 6:

$$\tilde{f}(\tilde{\hat{\mathbf{z}}}(\tilde{\nabla}_{\tilde{\hat{\mathbf{z}}}} \cdot \tilde{\mathbf{u}}_h) - (\tilde{\hat{\mathbf{z}}} \cdot \tilde{\nabla}_{\tilde{\hat{\mathbf{z}}}})\tilde{\mathbf{u}}_h + (\tilde{\mathbf{u}}_h \cdot \tilde{\nabla}_{\tilde{\hat{\mathbf{z}}}})\tilde{\hat{\mathbf{z}}} - (\tilde{\nabla}_{\tilde{\hat{\mathbf{z}}}} \cdot \tilde{\hat{\mathbf{z}}})\tilde{\mathbf{u}}_h) + \tilde{\nabla}_{\tilde{\hat{\mathbf{z}}}} \times (\frac{1}{\rho}\tilde{\nabla}_{\tilde{\hat{\mathbf{z}}}} p) = \vec{0} \tag{9}$$

Where we know $\tilde{\nabla}_{\tilde{\hat{\mathbf{z}}}} \times (\tilde{\nabla}_{\tilde{\hat{\mathbf{z}}}} p) = 0$ and $\tilde{\nabla}_{\tilde{\hat{\mathbf{z}}}} \cdot \tilde{\hat{\mathbf{z}}} = 0$ by definition. Water, being incompressible gives that $\tilde{\nabla}_{\tilde{\hat{\mathbf{z}}}} \tilde{\mathbf{u}}_h = 0$. Further assuming that our fluid is inhomogenous, meaning it has the same density everywhere, $\tilde{\nabla}_{\tilde{\hat{\mathbf{z}}}} \rho = 0$. Then, the equation reduces to:

$$\frac{\partial \tilde{\mathbf{u}}_h}{\partial z} = 0 \tag{10}$$

The local horizontal gradient should be constant across the entire depth of the fluid, a so-called 'stiffening effect'.

2.4 Thermal Wind

In a rotating, ideal gas, there are three balances when working in the rotating frame:

$$f u_y = -\frac{1}{\rho}\frac{\partial p}{\partial x} \qquad \text{geostrophic balance} \tag{11}$$

$$g = -\frac{1}{\rho}\frac{\partial p}{\partial z} \qquad \text{hydrostatic balance} \tag{12}$$

$$\rho = \frac{p}{RT} \qquad \text{ideal gas law} \tag{13}$$

where f is the Coriolis parameter, and is equal to 2Ω, where Ω is the frequency of the rotation. Equation 11 is a balanced flow from the primative equations, a common momentum balance used in atmospheric dynamics. We can compute,

$$\frac{\partial u_y}{\partial z} = -\frac{g}{fT}\frac{\partial T}{\partial x}. \tag{14}$$

However, for our fluid (water), the ideal gas law no longer applies. Instead, we need to use the **Boussinesq approximation**, given by

$$\rho = \rho_0 - \alpha\rho_0(T - T_0). \tag{15}$$

That is, the purpose of this approximation is to allow for buoyancy caused by temperature differences, while still maintaining a constant density. Note that α is the coefficient of thermal expansion. Using this instead of the ideal gas law, we can write

$$\frac{\partial u_y}{\partial z} = -\frac{\alpha g}{f}\frac{\partial T}{\partial x} \tag{16}$$

2.5 Baroclinic Instability

Baroclinic instability is a phenomenon by which thermal wind breaks down with fast enough background vorticity. This instability is responsible for the establishment of the characteristic 'eddies' (swirling patterns) seen in the mid-latitudes on Earth, central to the meridional transport of energy and weather. Thermodynamically, when surfaces of constant pressure and surfaces of constant density deviate, fluids can isentropically move to buoyantly unstable states and thus generate eddying motion. Climate physicists refer to the kinetic potential stored in the deviation of these fields as *Available Potential Energy* (APE) [5]. Analysis shows that the length scale of these eddies are given by a deformation radius:

$$L_\rho = \frac{\sqrt{gH}}{2\Omega} \tag{17}$$

with H as the scale height of the water. Eddies grow exponentially with an *Eady growth rate* given by:

$$\tau_\sigma = \frac{L_\rho}{0.31 \cdot U} \tag{18}$$

where U is the characteristic velocity of the flow. For a detailed analysis, see: [2].

Finally, the ratio between APE and kintic energy (KE) can be derived as in [5] to be:

$$\frac{APE}{KE} = \frac{2}{3}(\frac{L}{L_\rho})^2 \tag{19}$$

2.6 Important Lab Scales

It may be noted that equation 1 is written assuming that atmospheric air is *inviscid* or without viscosity. Since our experiments will be using water as a proxy for air, there may be specific cases where the viscosity of water plays an important role in governing flows. The *Ekman boundary layer* is a length scale around fixed surfaces where the viscous forces of a fluid balances the Coriolis force. This length scale, by [8] is approximated as:

$$\delta_{Ek} = \sqrt{\frac{\nu}{\Omega}} \tag{20}$$

Where ν is the kinematic viscosity of the fluid and Ω is the angular velocity. Correspondingly the associated timescale for spin-up given by the timescale can be argued by scaling to be:

$$\tau \approx \frac{H}{\sqrt{\nu\Omega}} \tag{21}$$

with H as the scale height of the water [1].

3 Particle Image Velocimetry (PIV)

Particle Image Velocimetry (PIV) is a technique used to measure the instantaneous velocity field within a fluid flow. It works by seeding the fluid with tiny tracer particles that are assumed to represent the fluid's motion. This section will describe PIV as a whole—see section 5.2 for detailed instructions on how to perform PIV analysis for this experiment.

For PIV to work effectively, the tracer particles must possess specific properties: they need to be small and light enough to accurately follow the fluid's streamlines and fluctuations without significantly lagging due to inertia, and they must also have neutral buoyancy or density close to that of the fluid. They must also efficiently scatter light from the illumination source towards the camera. Typical tracer particles in water have a diameter in the order of 10 µm.

The process of introducing these particles into the flow is known as seeding. PIV can also be used for finding air flows with special equipment being required for seeding. In liquids, particles are often pre-mixed into the fluid before the experiment begins. The amount of seeding, or particle density, is crucial for successful PIV measurements. If the seeding is too sparse, there may not be enough particle images within each computational interrogation window to produce a strong correlation signal, leading to inaccurate or missing velocity vectors. Conversely, if the seeding is too dense, particle images may overlap excessively, reducing image contrast, potentially attenuating the laser light significantly as it passes through the flow, and even potentially altering the fluid's properties. For intructions on seeding the experimental fluid, refer to Section 7.2.

To approximate the instantaneous velocity, you need track the data at two different times in quick succession. This can be done by using a pulsing laser that can emit two pulses, or taking two consecutive frames from a camera using burst mode or video capture.

4 Experimental Setup

This section describes the components and systems used for the Weather in a Tank experiments. Figure 1 gives a visual of the components involved (Figure 2).

4.1 Spinning Tank

The core of the setup is a cylindrical tank filled with water, placed onto a linear actuator (scissor lift) which itself sits on a rotating turntable. The rotating tank proxies the Coriolis effect present on Earth, which causes fluids (like air and water) to veer clockwise in the Northern Hemisphere and

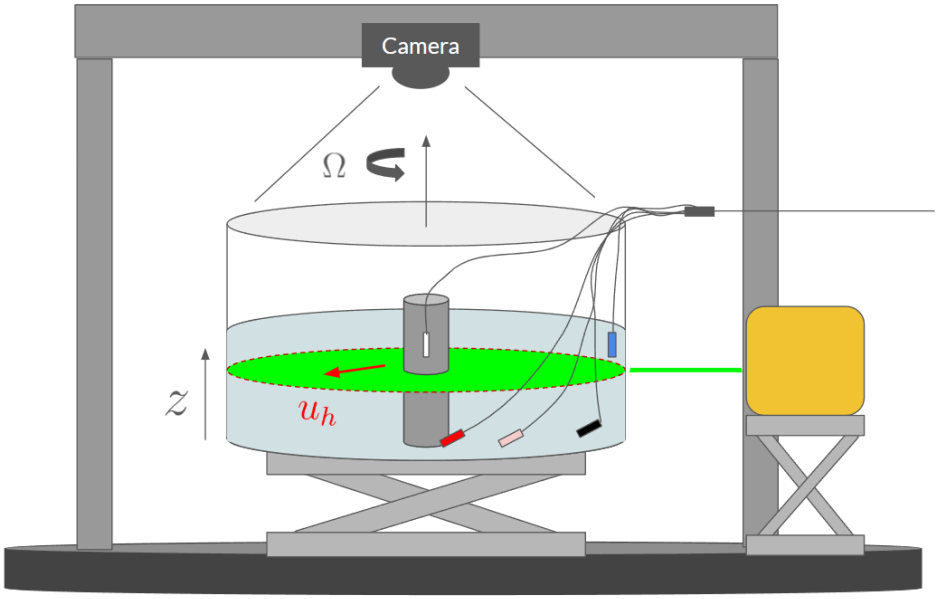


Figure 1: A diagram of the experimental setup. The cylindrical tank contains the water column used for experimentation (Section 4.1). The water will be seeded with silver-coated particles to scatter light for imaging (Section 4.2). A horizontal sheet laser is mounted on a scissor lift (Section 4.3) to the tank with an overhead camera to take the data (Section 4.4). Both the tank and the laser are mounted on scissor lifts so data can be collected at various relative heights. The entire setup is placed atop a turntable for adjustable velocity. A bottle is placed at the centre of the tank to simulate a temperature gradient, with various colour probes to monitor live temperature data at different points in the tank (Section 4.5).

counterclockwise in the Southern Hemisphere. The turntable is a Vue More variable speed model, capable of rotation speeds from 0.27 to 13.3 RPM. The entire system is enclosed within a large cardboard box to minimize external light, allowing for better visualization of flow tracers.

The scissor lift allows for vertical adjustments of the tank relative to the fixed laser sheet height. A key limitation is described in the Procedure.

4.2 PIV Setup

A planar laser sheet is used to illuminate a thin cross-section of the tank at a specific height. The movement of the PIV particles within this illuminated plane is recorded by a camera, allowing the fluid velocity field at that level to be determined using specialized software. Ensure particles are adequately dispersed before starting measurements. Note that the tank water must be stirred at regular intervals to return clumped PIV particles from the bottom of the tank back into the flow stream. For a reference on PIV, see [6].

Since there are a few parameters to play around with, it may be advantageous to try and observe the qualitative behaviour of flow you expect with dye visualization before you invest the time to setting up and calibrating the PIV apparatus. A smaller, 9 inch diameter glass tank is available to you at the station for preliminary tests.

4.3 Laser Illumination

A PREXISO PLC10UG cross-line laser leveler provides the illumination sheet. Tape is used to block the vertical portion of the laser cross, leaving only a horizontal line. The laser is mounted on a separate linear actuator for height adjustment and should use one the provided 3-D printed laser leveling wedges to ensure the laser sheet remains horizontal and stable throughout the experiment. The different labeled wedges are shown in Table 1, with the corresponding lower side thickness provided. For consistent results, place the laser on a flat surface, allow it to auto-level, then lock its orientation (refer to the laser manual). The flatness of the laser can be checked by shining the laser through the tank and checking to see if the laser height is the same on the other end after rotating the tank 180 degrees. See Figure 3 for an

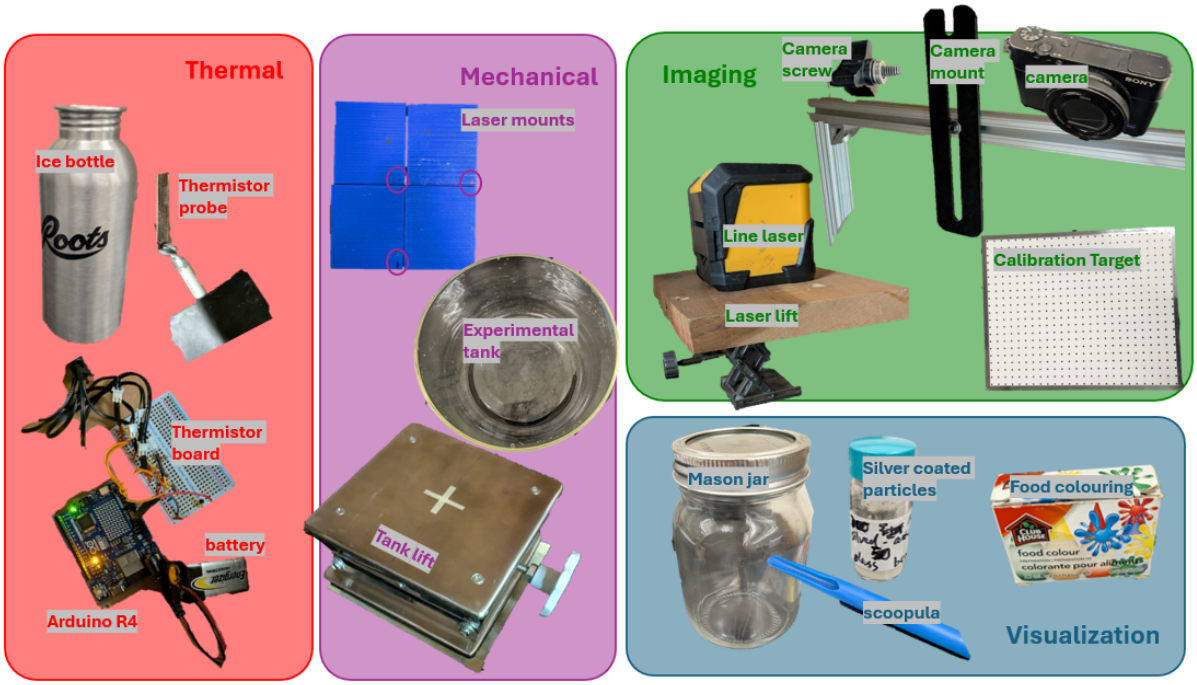


Figure 2: Experimental apparatus sorted by category: imaging, visualization, mechanical, and thermal. Food colouring is included as a qualitative test if you would like to verify the magnitude and direction of flow visually before taking quantitative data with PIV. Two tanks will be available to you, a larger, plastic tank for PIV tests and a smaller glass tank which you may use for preliminary testing.

example. If you find that it is difficult to read laser sheet level due to how diffuse the signal is, it may be time to clean the tank (Section 7.1).

Wedge Label	Angle (°)
1	2
2	2.5
3	3
4	3.5

Table 1: Wedges with corresponding angles, their label is written on the bottom right corner as circled in purple in Figure 2.

Warning

Laser Safety: This is a Class 1 laser. While generally safe under normal operating conditions, avoid staring directly into the laser beam. Reference [3] for details.

4.4 Camera and Imaging

A camera is used to record the PIV particle movement. It can be screwed onto a mount inside the cardboard box enclosure. The mount allows for vertical positioning and horizontal shifting.

- **Focusing:** Access the camera's settings and select "MF" (Manual Focus). Use the menu controls (often left/right buttons) to adjust the focus. For PIV, place the calibration target in the tank, aligned with the laser sheet. Adjust focus until the target dots are sharp. Take a clear photo of the focused target for software calibration, then remove the target. Re-check focus on the PIV particles themselves once seeded and illuminated. A focused camera on the particles is absolutely critical for the functionality of the PIV algorithm and the focus should be checked and/or adjusted regularly. Examples of focuses can be seen in Figure 4.

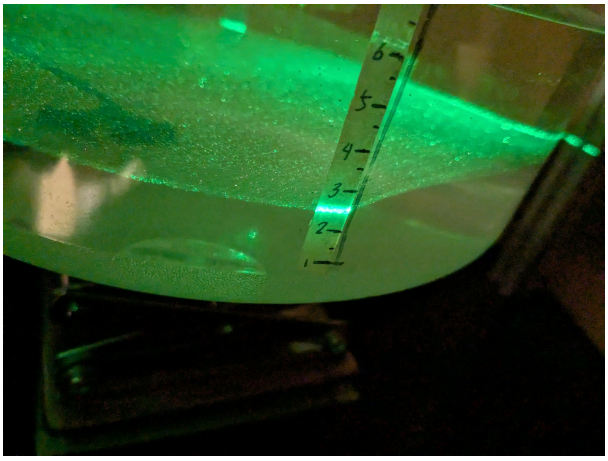


Figure 3: An example image of what the laser sheet looks like illuminating the ruled markings on the tank.

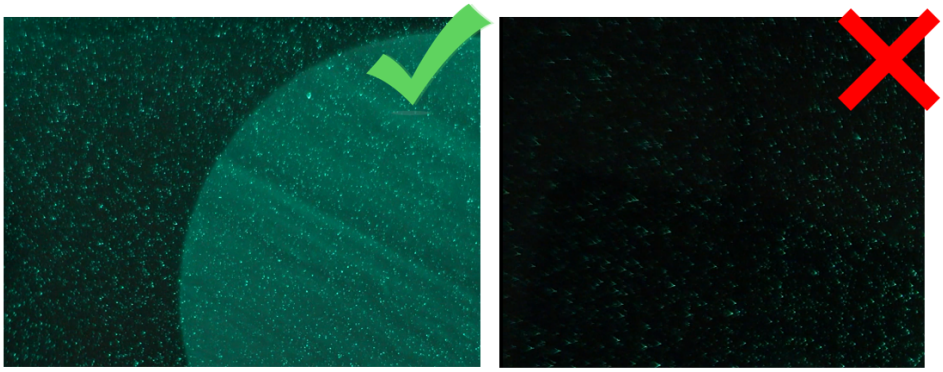


Figure 4: Example of good focus on the left, dense seeding of PIV particles allows the dots to be imaged sharply. The size and shape of particles are uniform across the region. Example of poorly focused image on the right, the region is dark, with only some particles visible. Not uniform acorss the image suggesting that the focal plane is misaligned relative to the laser plane.

- **Data Transfer:** Images and videos are stored on an SD card within the camera, which can be connected to the lab computer.
- **Power:** Note that the camera battery drains relatively quickly and will likely need frequent recharging between or during experimental runs. Plan accordingly.

4.5 Temperature Measurement System

In-tank temperature gradients are measured in real-time using thermistors connected to an Arduino micro-controller equipped with a WiFi module.

- **Probes:** The thermistor probes are colour-coded for identification. Position them carefully as needed for the specific experiment. See Table 2 for color codes and example locations.
- **Data Acquisition:** The Arduino automatically uploads temperature readings via WiFi to the Arduino Cloud approximately every 15 seconds. Ensure the lab's WiFi is connected.
- **Accessing Data:** Log in to the Arduino Cloud using the provided credentials:
  - **Email:** apl.atmos@gmail.com
  - **Password:** APLatmos13579

  Navigate to the "Dashboards" tab and select the "Temp-display" dashboard to view live data (see Figure 5).

- **Exporting Data:** Click the export button on the dashboard view, select the desired probes (T0-T4), and the timeseries data will be emailed to *apl.atmos@gmail.com* as a downloadable .CSV file. Make note of the time on the Arduino and the time on the camera for correspondences
- **Configuration:** Modifications to the Arduino code (e.g., changing the sampling interval) are done via the cloud interface ("Things" tab/"Temp_Collect" project/ Sketch). Uploading changes requires connecting the Arduino to the lab computer via USB-C and clicking 'upload' in the cloud editor. This connection is generally only needed for configuration, not regular data logging.

Table 2: Thermistor Color Coding and Example Placement

Color	Label / Example Location
Red	T0 (Near water bottle, lower level)
Black	T1 (Near tank edge, bottom level)
Pink	T2 (Mid-radius, bottom level)
Blue	T3 (Near tank edge, top level)
White	T4 (Reference - e.g., inside ice bottle)

Note: These locations are examples; position them carefully as needed for the specific experiment, avoiding the main PIV imaging area if possible.

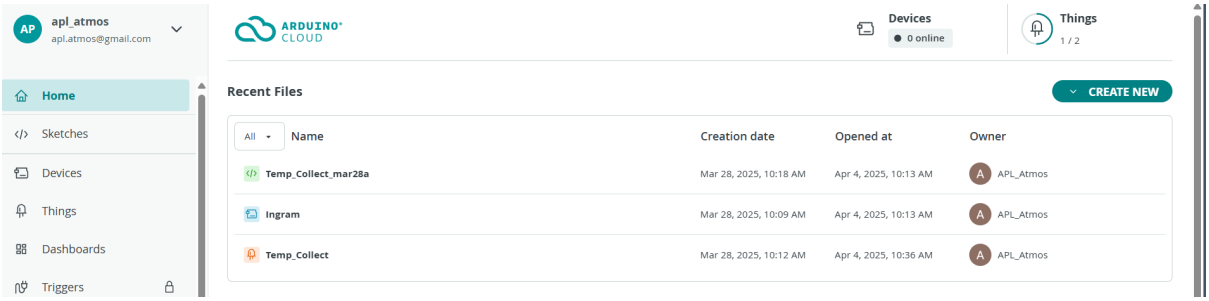


Figure 5: The Arduino Cloud homescreen interface.

Warning

Thermistor Connection Fragility: The thermistor wire connections to the Arduino or breadboard are delicate. Handle them with extreme care during setup, placement, and removal. Avoid pulling on the wires. If connections fail, resoldering might be necessary. Consult the provided circuit diagram (Figure 6) or lab staff if issues arise.

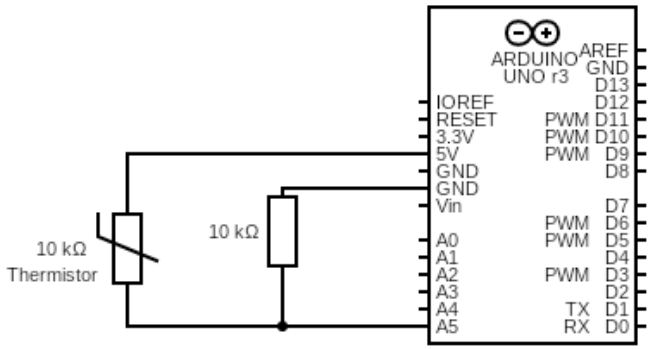


Figure 6: Example circuit diagram for the thermistor connection to input A5. All other thermistors are connected identically.

4.6 OpenPIV Software

Data analysis, specifically the calculation of velocity fields from the recorded images/videos, is performed using routines based on the Open PIV project, typically via provided Python scripts [4].

5 Procedure

This section details the steps for conducting experiments.

5.1 General Preparation Steps

For safety and equipment protection please review these steps carefully before starting the rotation and measurements.

1. **Align Tank and Laser:**
   - Place the empty tank on the scissor lift turntable. Use the tape marker for consistent centering.
   - Set up the laser on its actuator/mount. Ensure it is level.
   - **Height Planning:** Position the tank using the scissor lift so the laser sheet illuminates the *lowest* planned measurement height inside the tank. Note this initial height.

     Warning

     Tank Actuator Limitation: The scissor lift actuator mechanism cannot safely raise the tank's height while the tank is filled. It is designed only to lower the height safely under load. Plan your experiment starting with the laser aligned to the **lowest** desired level. Subsequent measurements at higher levels are achieved by **lowering** the tank relative to the fixed laser sheet.
   - Test that you can see the PIV particles with the laser. If not, you may need to gently stir them.
2. **Temperature Measurement:** *For Experiments 6.3 and 6.4 only. Skip these steps if your experiment does not require a temperature gradient.*
   - **Connect Arduino and Thermistors:**
     - Connect the Arduino to power. Ensure it connects to the lab WiFi network.
     - Log in to Arduino Cloud on the lab computer and verify that the "Temp-display" dashboard is receiving data. Troubleshoot connection if necessary. Check battery level if using battery power.
   - **Position Thermistors:**
     - Using waterproof tape, securely attach the thermistor probes inside the tank according to your experimental plan (refer to Table 2 for color codes and examples).
     - Route the wires carefully over the tank edge, securing them so they do not interfere with rotation or PIV measurements.
   - **Prepare Ice Source:**
     - Fill the metal bottle with crushed ice, ensuring the stabilizing weight is inside. Pack tightly.
     - Carefully place the bottle in the center of the tank.
3. **Set up Camera:**
   - Mount the camera securely onto its designated mount inside the enclosure. Point it downwards towards the tank.

Warning

Camera Mounting Hazard: The 1/4"-20 mounting screw can hold the camera securely *if tightened properly*. Double-check the tightness before letting go. Be extremely careful when mounting, adjusting, and removing the camera to avoid dropping it into the water tank. Work slowly, **keep the camera wrist strap on**, and maintain a grip on the camera until secure.

- Adjust camera position (vertical/horizontal) and focus (Manual Focus - MF) until the calibration target dots within the laser sheet are sharp and fill the desired field of view.

4. **Final Pre-Spin Checks:**
   - Double-check that all wires are clear of the turntable and will not snag during rotation.
   - Confirm secure mounting of the tank, laser, camera, and ice bottle.
   - Verify water level and PIV particle distribution.
   - Note the time.
5. **Start Rotation and Achieve Equilibrium:**
   - Turn on the turntable. Adjust the rotation speed to the target rate (e.g., **1 RPM**).
   - **Wait for Equilibrium:** *for thermal gradient experiments only*
     Allow sufficient time for the initial transients (spin-up) to dissipate and for the thermal gradient to establish steady flow. This may take several minutes. One way to quantify this is to first do a control measurement without a temperature gradient, and quantify how long that takes to reach equilibrium. Note that with a temperature gradient, the system can be considered near equilibrium when the large-scale flow patterns appear relatively stable and temperature fluctuations become statistically steady.

Challenge

Quantify the temperature fluctuations (e.g., calculate the standard deviation of readings from T0-T3 over rolling time windows) to objectively determine when a statistically steady thermal state is reached. Compare the time taken to reach this state with theoretical estimates for spin-up time (e.g., Ekman timescale) and thermal diffusion time.

6. **Record PIV Data at Multiple Heights:**
   - Ensure the laser is on and illuminating the PIV particles at the current (lowest) height.
   - Start camera video recording. Ensure you have a method to correlate time in the video with the Arduino temperature data timestamps (e.g., note the start time).
   - Record data at the current height.
   - **Adjust Height:** Carefully and slowly turn the scissor lift knob to *lower* the tank to the next desired measurement height (e.g., increment by 1 cm). Note the new height precisely.

     Warning

     Adjusting height while rotating requires care. The movement can disturb the flow or cause spills if done too quickly or unevenly. Ensure the adjustment is smooth to minimize perturbations.

   - Allow a brief moment for the flow at the new level to stabilize after adjustment if needed.
   - Record video at the new height for the set duration.
   - Repeat the height adjustment and recording process for all planned heights.
7. **Save Data:**
   - Once you are finished collecting data, stop the camera video recording.
   - Slowly and smoothly decrease the turntable rotation speed to zero. Turn off all equipment.

- *For temperature measurements*: log in to Arduino Cloud and export the temperature data from the relevant time period. Select probes T0-T4 and ensure the data is emailed to *apl.atmos@gmail.com*. Download the .CSV file from the email and save it with a descriptive name.
- Carefully remove the camera from its mount. Connect the camera or SD card to the lab computer and transfer all video files and the calibration photo(s). Save them with descriptive names indicating the experiment and height level.
- *For temperature measurements*: carefully remove the thermistor probes and the ice bottle.
- Begin cleanup procedures

5.2 PIV Processing and Data Analysis

This section outlines how to process the recorded videos into quantitative velocity field data using our provided PIV software. The 3-stage processing software should be available on the lab computer, and the source code is available on GitHub. If you have any questions, bug reports, or would like to contribute feel free to use the issues page or email me. I

5.2.1 Preprocessing

Use the video editor to trim and crop raw recordings before PIV analysis.

1. **Launch the Tool:** Run the Python script for the visual video editor.
2. **Load Video:** Select a raw `.mp4` file from the experiment.
3. **Load Calibration Image:** (opt.) Select a `.jpg` file of the calibration frame. The software will **automatically** process the calibration image with the appropriate settings based on what you did to the experiment video.
4. **Trim Time Segment:** Use the timeline slider to define a start and end time after flow equilibration. Set these using the buttons or by typing directly.
5. **Crop Region:** Adjust the red rectangle over the laser-illuminated area containing visible PIV particles. Do this visually or by entering `X1`, `Y1`, `X2`, `Y2` coordinates. We recommend using the same crop coordinates for all heights in the same experiment.
6. **Save:** Click `Process & Save Video`.

5.2.2 Calibration

(optional) Use the calibration engine to perform calibration and obtain an approximation of the camera-to-world (C2W) map. Your settings are all on the top-left panel of the program.

1. **Load Image:** Select the pre-processed image from the previous stage. The image should appear three times.
2. **Enhance Frame:** Choose an appropriate window size, which should be similar or slightly smaller than the diameter of the circles of the calibration target (in pixels). You may also measure the circle by left-clicking and dragging on the top-left image. Once you are satisfied, press `Run Enhance`.
3. **Filter Components:** Click on `Run Filter Comp` to perform filtering with the default parameters. You should see the orange circles in the bottom-left image, which indicates the dots the algorithm identified. If you are not satisfied, you may change the parameters and click `Run Filter Comp` again to refresh the dots.
    - *Corr. Thresh* (default 0.3) is the minimum intensity in the enhanced frame for pixels to be considered part of the dots. Typically, if you see too few dots you should decrease this parameter. If you see too many erroneous dots, you may want to increase this parameter. **Warning:** if you decrease this parameter too much, it may interfere with the following parameter and decrease the number of dots.

- *Comp. StdDev Thresh* (default 2.0) is the threshold for removing outlier points based on width, height, and density of the dot candidate. Increasing this will filter out less points, thus, if you see too few circles you may want to decrease this parameter.

4. **Filter Centroids:** Click `Run Filter Cent.` to perform filtering with the default parameters. This filtering stage removes dots that are either too far apart of too close to its nearest neighbor. **Tip:** If you are struggling, leave this setting at 5.0 and filter points manually.
5. **Manual Filtering:** You may left click on the red circles in the botton-right image to manually filter away points.
6. **Run Calibration:** Clicking the `Run Calibration` button will perform the calibration, and display the world coordinates of the dots in the reprojection plot. It will also pop up the angle your calibration target is rotated. Make sure the reprojection plot matches your target once you rotate your perspective.
7. **Save:** Click `Save Parameters` to save the calibration parameters into a `.npz` file: (the intrinsic matrix K, distortion vector d, extrinsic matrix $[R|t]$ and the rotation angle θ.

5.2.3 PIV Analysis

Use the GUI tool to compute velocity fields from processed videos.

1. **Load Video:** Select a pre-processed video.
2. **Set Parameters:** Define `Window Size`, `Overlap`, `Search Area`, `S/N Threshold`, `Displacement (dt)`, and frame range. For the first time, we recommend using the default parameters. You may click on the question mark icon next to the parameters to learn more about them. If you are more curious, you may refer to the source code of `extended_search_area_piv`.
3. **Calibration:**
   - (If calibration is performed) Select the calibration parameters that was saved as `.npz`
   - (If calibration is skipped, for testing) Manually define a scale factor in mm per pixel.
4. **Run and Repeat:** Click `Run PIV Analysis`. You may see the and process all other height-level clips the same way.

5.3 Plotting and Output Files

- `..._world_data.csv`: Contains frame index (i.e. time), position (x,y), and velocity components (u, v) in mm. It also includes the signal to noise ratio.
- `..._summary.txt`: Summary of input settings, time step Δt, and mean/RMS velocity magnitudes.

The file `..._world_data.csv` contains all the data extracted from the PIV process, and you are free to perform your own analysis and make your own plots. As a starter, `plots.py` will provide some basic plots.

5.4 Clean Up

Once you are finished with experiments for the day, ensure that all you data has been saved appropriately. The tabletop should remain plugged in and on but *not spinning*. Ensure that all other electronics such as the arduino, laser, and camera are fully turned off. It may be wise to leave the camera battery charging for the next lab session. If you are at the end of your APL sessions for this rotation, please empty and clean the tank according to Section 7.1.

6 Experiments

The following set of experiments provide some guiding questions for initial exploration with the rotating tank apparatus. You may find inspiration from similar lab experiments through MIT's Weather in a Tank Wiki.

6.1 Solid Body Rotation: Inertial Flows

This experiment is designed around the inertial flows observed over oceans (Equation 8) which is an example of solid body rotation (SBR). Temperature gradients are not required. Follow General Preparation Steps to set up the station and acquire data to answer the following questions:

6.1.1 Questions

- Do you see a radial pattern as described by Equation 8? Is there any variation in horizontal velocity as you move up the water column? Does this hold for different background vorticities?
- What is the shape of the free surface of the water when it is spinning? Hint: how can you find constant pressure surfaces with the velocity field?
- Starting from the stationary case, how long does it take to reach SBR? Is this a function of the background vorticity?
- How well does the theory hold up near the surface of the water, near the bottom of the tank, and near the walls on the edge? Are you able to quantify a length scale around the boundaries where the theory no longer holds? Does this vary with background vorticity?

6.2 Taylor Column Experiment

This experiment investigates the behaviour of the flow predicted in Equation 10 by placing a cylindrical barrier at the bottom of the tank and imaging the horizontal flow field at various heights both at the level of the barrier and above it. Follow the General Preparation Steps. Once you are ready to spin, set the turntable to an intermediate velocity, around 5 RPM. Wait until solid body rotation is achieved by checking that particles look stationary in the camera data. Begin recording and change the rotation slightly, by around 0.5 RPM. You may find that taking a video recording of around 30 seconds in length after the rotation adjustment is sufficient.

6.2.1 Questions

- How does both the magnitude and direction of flow around the barrier vary with height? Does this match expected theory?
- How does the behaviour hold for various background vorticities? Is there a threshold background vorticity and difference in RPM after which the theory is not longer valid?
- how does 'stiffening' hold up to different barrier shapes?
- **fun bonus:** can you, through both theory and empirical testing, produce an algorithm to 'piece together' the shape of an arbitrary barrier by imaging the flow field above it?

6.3 Thermal Wind Experiment

In the Thermal Wind experiment, a thermal gradient is introduced, typically by placing a cold source (like an ice-filled bottle) in the center of the tank and allowing the outer edge to represent a warmer region. This proxies the temperature difference between Earth's poles and equator. The tank now mimics a *baroclinic* atmosphere, where surfaces of constant temperature (isotherms) are no longer parallel to surfaces of constant pressure (isobars). The tendency for the fluid to equilibrate this temperature difference is counteracted by the Coriolis effect (due to rotation), leading to a vertical shear in the azimuthal velocity known as the thermal wind. Measuring the velocity field at different heights is crucial for observing this effect.

6.3.1 Questions

- From your PIV results for the Thermal Wind experiment, plot the mean azimuthal velocity as a function of radius for each measured height. Does the shear (change in velocity with height) match the direction predicted by the thermal wind balance equation?

- Compare the measured temperature difference between the center and edge with the observed vertical shear in azimuthal velocity. Can you estimate the thermal wind shear quantitatively and compare it to theory?
- How did the observed thermal equilibration time compare to the Ekman spin-up time calculated for your experimental parameters? Discuss any discrepancies.

6.4 Eddies and Mid-latitude Jets

This experiment investigates the onset of *baroclinic instability* which manifests itself in the characteristic cyclonic and anticyclonic 'eddies' seen in the mid-latitudes on Earth. Baroclinic instability happens when the lines of constant density in a flow field deviate from the lines of constant pressure, creating a source of potential energy for the generation of these eddies. In this experiment, you may use differential heating with an ice bath, along with the rotation of the tank to produce two distinct pressure and density fields.

6.4.1 Questions

- For which temperatures and rotation speeds do you see the onset of baroclinic eddies?
- How do the size and count of baroclinic eddies change with rotation speed and temperature gradients?
- What is the timescale of formation for these eddies? How long do they last once generated?
- How much kinetic energy is carried through the eddies? Do you see a correspondence between energy dissipation by eddies and the transient temperature data?

7 Maintenance

7.1 Cleaning the Tank

Dust accumulation and various debris can degrade the signal and performance of the PIV system. Periodically (perhaps once a month), it may be necessary to dump the liquid and clean the tank.

Warning

The PIV particles are around 10 µm large and improper disposal can pose a threat to the environment and ecosystem health.

To properly dispose of the liquid, simply use the provided funnel to empty out the tank into the hazardous waste disposal tank under the table. The tank may be heavy (up to 5 lbs or more) so you may require another person to help you lift and move it. Be careful to empty the tank slowly as to avoid spills and splashes.

After you empty out the tank, there may be residue left at the bottom. You may use a wet paper towel or sponge to sequester the remaining particles for disposal.

7.2 Seeding the Water

If you need to add more PIV particles to your water, please use the provided plastic scoopula (Figure 2) to portion out a small amount into a water-filled mason jar and mixing, before introducing it into the larger tank. This prevents clumping and ensures more even distribution of seeding throughout the tank.

Only a very small amount of silver-coated particles is necessary for visualization. For example, Figure 7 shows how much is necessary for the entire tank, scale accordingly to your needs. A good way to know whether your seeding is sufficient is if you see a 'starry night' when you shine the laser through the seeded water, with a visual density similar to the good example in Figure 4.



Figure 7: An example of the quantity of particles necessary for running experiments in the provided tank.

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